5 V SMPS – Professional One-Night Layout & Validation Guide

Objective

You are not simply creating a functional power supply. The goal is to produce a PCB layout that clearly demonstrates power-integrity awareness, EMI discipline, and validation readiness. This document outlines a focused, professional workflow suitable for rapid execution while maintaining industry credibility.

Scope Lock (Do Not Expand)

- Single-output 5 V SMPS
- Target power: 20–30 W
- No additional monitoring, interfaces, or experimental features
- Priority: clean layout, measured results, documented intent

Finishing a clean, measured design carries more professional weight than expanding scope.

Placement Order (Critical)

1. Switching Loop Placement
Place the controller, primary switch, snubber, bulk capacitors, and transformer/inductor first. Minimize the high di/dt loop area.

2. Secondary Rectification Loop
Place rectifier diode and output capacitors tightly together to minimize current loop area.

3. Control & Feedback Network
Route feedback from the output sense point. Keep feedback away from the switch node. Provide a quiet local ground reference.

4. Input and Output Connectors
Ensure direct current paths from connectors to bulk/output capacitors with no unnecessary detours.

Grounding Strategy

Recommended approach: Solid ground plane on Layer 2.

- Keep high-current returns local
- Connect control/analog ground at the controller ground pin
- Avoid aggressive split planes unless required

Clean, intentional grounding is preferred over clever complexity.

Layer Stack Recommendation

For a 4-layer board:
- L1: Power and signals
- L2: Solid ground plane
- L3: Power pours (VIN, VOUT)
- L4: Signals and low-current routing

For 2-layer designs, use wide pours and aggressive ground stitching.

Routing Rules (Non-Negotiable)

- Keep switch node copper as small as possible
- Short, direct gate-drive routing
- Place snubber components directly at device pins
- Route feedback traces as sensitive signals, not power

Violations should be corrected before ordering.

Thermal Design Intent

- Copper pours under hot components
- Thermal vias where possible
- Avoid routing sensitive signals under hot switching nodes

Even without simulation, visible thermal intent matters.

Silkscreen & Documentation

- Clearly label VIN and VOUT polarity
- Label test points
- Include revision ID (e.g., 5V_SMPS_REV_A)
- Include date or initials

These details quietly signal professionalism.

Validation Checklist

After assembly, measure and document:
- Output voltage vs load
- Efficiency vs load
- Output ripple
- Load transient response
- Thermal behavior

Measured performance is what converts a design into a portfolio asset.

Portfolio Capture

Before ordering and after bring-up, capture:
- Overall PCB view
- Switching loop zoom
- Feedback routing zoom
- Layer stack view

These visuals support interviews and applications.

Professional Rationale

A clean, measured 20–30 W SMPS demonstrates judgment, restraint, and execution discipline. These traits are valued more than raw power capability in entry-level power and PCB roles.